



Landscape and functional groups moderate the effect of diversified farming on biodiversity: A global meta-analysis

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ABSTRACT

Homogenisation and intensification of agricultural ecosystems are among the most important threats to biodiversity, linked to declines in pollinators, soil biota and ecosystem functioning. Diversification has been proposed as a way to restore ecosystem functioning in agricultural landscapes. To manage agricultural land for multiple ecosystem functions, evidence is needed of the effect of diversification on functionally distinct taxa. We contribute to closing this knowledge gap through a global meta-analysis of 161 peer-reviewed articles addressing the abundance and richness of six distinct functional groups: autotrophs, decomposers, natural enemies, pests, pollinators, and other. We found diversified farming systems increased overall species richness by 26% on average, relative to simplified farming systems. However, the effect of diversified farming on the overall mean abundance was weak. Our study shows diversified farming systems enhanced the abundance and richness of beneficial species while reducing the abundance of pests (e.g., weeds, herbivores), thus providing benefits for both agricultural production and biodiversity. The positive effect of diversified farming systems on the overall mean species richness was stronger in farms in more simplified landscapes, i.e., those that are further from natural and semi-natural habitats or have a lower proportion of seminatural vegetation in a 1 km radius. Pollinator's abundance and richness were highest in diversified farming plots located far away from natural and semi-natural habitats. In contrast, proximity to these natural and semi-natural habitats (<250 m) increased the positive effect of diversified farming systems on natural enemies' abundance, while reducing the number of pests. Our results add to the body of evidence calling for the repurposing of policies, regulations, and international agendas to promote, support and incentivize the adoption of diversified farming practices for supporting biodiversity. Spatial planning of diversification schemes should consider the landscape context of farms to ensure the greatest benefit of intervention.

1. Introduction

The last century has seen traditional farming gradually displaced across the world by conventional systems to increase food production for the world's growing population (Krebs et al., 1999; FAO et al., 2019; UN, 2016). Although the expansion and intensification of agriculture that is associated with conventional management has increased food production (Ramankutty et al., 2018), it also constitutes the most important threat to biodiversity (Maxwell et al., 2016).

Conventional farming has resulted in the simplification of agricultural mosaics at multiple scales (spatial and temporal) and levels (farm and landscape). At the farm level, intensification has driven agricultural production to specialise on a few crop species (e.g., rice, maize, cotton) with high dependence on external inputs (e.g., irrigation, fertilisers, and pesticides) (FAO, 2017). Moreover, high-intensity agriculture has also eliminated or shortened crop rotation cycles (i.e., only 1- or 2-crop rotation cycles) (Plourde et al., 2013). At the landscape level, the expansion of conventional agriculture has caused the widespread

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conversion of natural systems into expansive monocultures (Laurance et al., 2014).

This homogenisation and intensification of agricultural systems has reduced the abundance and richness of species that benefit from habitat heterogeneity (Benton et al., 2003; Donald et al., 2001; Gonthier et al., 2014). For example, Woźniaka and Kawecka-Radomska (2016), found that in cereal monocultures the number of earthworms decreased as a result of the scarce and homogeneous organic matter. The over exploitation of agricultural ecosystems contributes greatly to biodiversity loss, and therefore, a decline in ecosystem services provision such as pest suppression, pollination, nutrient cycling, and water regulation (Kremen, 2020; Tamburini et al., 2020). Such reductions are putting at risk agroecosystem sustainability, resilience and their capacity to meet food production demands (FAO et al., 2019).

Diversification in agricultural systems has been proposed as a strategy to ensure both food security and environmental sustainability (Maggio et al., 2018). Diversified farming differs from conventional monoculture by associating different varieties and/or species of plants, fish, and/or livestock, at multiple temporal and/or spatial scales (Kremen et al., 2012). By increasing temporal and spatial heterogeneity, diversified farming should help restore and stabilise populations of non-domesticated species, including key providers of ecosystem services. Studying how functionally diverse species vary under different agricultural systems can provide insights into how to maintain associated ecosystem functioning and services.

Several studies (e.g., Bagayoko et al., 2000; Woźniaka and Kawecka-Radomska, 2016) have found that temporal diversification – such as crop rotations – increases the abundance and richness of nutrient recycling species, thus improving soil fertility. Evidence also suggests that plant diversification within plots – such as intercropping and agroforestry – provides shelter and alternative food sources for natural enemies of pests while reducing the abundance of crop pests (Barrios et al., 2018; Gontijo, 2019; Snyder, 2019).

At the landscape scale, the maintenance of natural and semi-natural habitats around and between agricultural fields can promote the diversity of species that benefit crop production (e.g. pollinators, natural predators of crop pests), by providing shelter, food, nesting sites and more stable micro-climates (Garibaldi et al., 2014; Gontijo, 2019). Agricultural landscapes with greater forest cover support higher numbers of bird and bat insectivores, increasing crop pest predation rates (Karp et al., 2013), while proximity to natural and semi-natural areas positively influences the diversity of pollinators in croplands (Garibaldi et al., 2011). Therefore, the effectiveness of local farming strategies to promote biodiversity may depend on the surrounding landscape context (Petit et al., 2020; Tschamntke et al., 2012). For example, the positive effect of diversified farming systems tends to be higher in more complex agricultural landscapes (Kennedy et al., 2013; Lichtenberg et al., 2017), while organic farms are more effective in very intensive landscapes (Feber et al., 2015; Tuck et al., 2014).

Several previous studies have synthesised the impacts of diversified farming systems on the abundance and richness of different functional groups of non-domesticated biodiversity. However, these studies are limited to a few taxa (mostly arthropods), regions, and/or farming system types (Kennedy et al., 2013; Letourneau et al., 2011; Lichtenberg et al., 2017; Shackelford et al., 2013). Previous meta-analyses have also assessed the influence of landscape complexity on non-domesticated biodiversity, for certain taxa and landscape complexity measures (Batáry et al., 2011; Dainese et al., 2019; Kennedy et al., 2013; Lichtenberg et al., 2017). Overall, these syntheses have showed that biodiversity tends to be higher in diversified farming systems surrounded by complex landscapes. A new more comprehensive meta-analysis is merited firstly because hundreds of primary studies emerge every year providing fresh evidence and previous meta-analyses incomplete. Moreover, previous studies that systematically synthesised the effect of diversified farming systems on different functional groups tend only to consider pollinators, pests or natural enemies (and often only one of

these) while decomposers and trade-offs between functional groups are understudied. Existing global reviews of effects on biodiversity rely on a relatively small number of effect sizes (< 100) (Lichtenberg et al., 2017; Shackelford et al., 2013), or use vote counting which does not reveal anything about the magnitude of an effect (Rosa-Schleich et al., 2019).

Here, we performed a global meta-analysis of 161 articles to assess the influence of diversified farming practices on the abundance and richness of non-domesticated terrestrial biodiversity. We classified species into functional groups important for ecosystem functioning where this can benefit agricultural production, focusing on pollination, pest control, and nutrient cycling functions. Additionally, we conducted a supervised land-cover classification around each study plot to obtain measures of landscape complexity at 1 km radius in terms of configuration (distance to natural/semi-natural habitat), heterogeneity (land cover Shannon's diversity index), and composition (proportion of natural/semi-natural habitat). Grounded in current evidence, we hypothesised that: (i) Diversified farming has an overall positive effect on the abundance and richness of non-domesticated biodiversity; (ii) Effects of diversified farming varies among functional groups, with autotrophs, decomposers, natural enemies of pests, and pollinators responding positively and crop pests responding negatively; (iii) Effects of diversified farming on the overall abundance and richness of non-domesticated taxa are more positive in complex landscapes.

2. Methods

2.1. Literature search

Jones et al. (2021) searched for primary studies published in peer-reviewed journals, comparing different outcomes of non-domesticated terrestrial biodiversity (e.g., abundance, species richness, Shannon's diversity index) between diversified farming systems (e.g., intercropping, agroforestry), and either simplified farming systems (e.g., monoculture) or natural habitats (e.g., forests, shrubland). The literature search was accomplished in three different stages, following the protocol described in Sánchez et al. (2021). The first involved the systematic search of relevant studies in Scopus and Web of Science using search strings without restrictions for the publication year or location of the studies (see search string in Table A.1). The second consisted of extracting the list of primary articles included in all meta-analyses found in stage one. Third, Jones et al. (2021) considered additional relevant primary studies suggested by consulted scientists and which were not picked up by the first and second stages. Jones et al. (2021), identified a total of 1545 unique articles with the potential to be included in their meta-data (Fig. A.1).

The abstract, methods, and results sections of all identified articles were carefully read to determine their eligibility for inclusion. Finally, Jones et al. (2021) meta-data contained 4076 comparisons from 237 articles (see Dataset 2).

2.2. Meta-data

For this meta-analysis, we only used comparisons of outcomes that met the following inclusion criteria: (i) biodiversity outcomes were provided as species richness (i.e., number of species sampled) or abundance (e.g., number of individuals sampled) in simplified farming systems (comparator) and diversified farming systems (intervention) (see Table A.2 for description of diversified and simplified systems); (ii) terrestrial biodiversity was assessed and could be classified into one of six functional groups (autotroph, decomposer, natural enemy, other, pest, pollinator); (iii) biodiversity means and standard deviations were greater than zero ($\mu > 0$) for both the intervention and comparator systems (necessary to calculate log response ratios); (iv) geographic coordinates or detailed location name of the study site were provided. Our dataset included 2258 comparisons (from 161 articles), of which 1886 (129 articles) measured abundance, and 372 (65 articles)

measured richness (see Fig. A.1).

We extracted the following qualitative and quantitative data for each included comparison (i.e., intervention and comparator): (i) article ID, (ii) taxonomic group (based on order and phylum), (iii) functional group, (iv) biodiversity outcomes (e.g., abundance, species richness), (v) system comparison class (i.e., simplified or diversified farming), (vi) location data; (vii) year of publication; (viii) internal validity assessment criterion (i.e., low, high or unknown risk of bias); (ix) biodiversity means (μ), (x) standard deviation values (SD); and (xi) number of samples (n).

2.3. Meta-data processing

We reclassified the functional groups into six classes (autotrophs, decomposers, natural enemies, pests, pollinators and others) based on the information provided in the original article and the direct or indirect ecological role that they play in improving or reducing crop yield and quality (see Table A.3). Taxa order and phylum were further reclassified into ten taxonomic groups (Amphibians, Annelida, Arthropoda, Eukaryotes, Birds, Fungi, Mammals, Nematoda, Plantae, Reptiles).

Each included comparison was classified as High, Low or Unknown risk of bias, for internal validity assessment (i.e., assessment of risk of bias). This approach aims to minimise risk of bias caused by the inclusion of comparisons from articles with poor quality methodologies (CEE, 2013; Mengist et al., 2020). We rated the comparisons considering three quality criteria: (i) intervention and comparator sample size (i.e., low risk of bias: sample size ≥ 5 ; high risk of bias: sample size < 5), (ii) intervention and comparator sites are within ~ 1 km (i.e., low risk of bias: same climate conditions, weather events, soil type; high risk of bias: different climate conditions, weather events, soil type); (iii) time the interventions and comparators plots have been in this state (i.e., low risk of bias: ≥ 1 year; high risk of bias: < 1 year; unclear risk of bias: insufficient data).

2.4. Landscape analysis

We quantified landscape characteristics in circular areas of 1 km radius (≈ 3.14 km²) around each study point (using geographic coordinates associated with each comparator/intervention as the centroid). We selected a 1 km radius since the abundance and diversity of multiple species and taxa are best predicted at this distance (e.g., Fijen et al., 2019; Heim et al., 2015; Rodewald and Yahner, 2001). We conducted a land cover supervised classification (i.e., collecting ≈ 30 samples across spectral reflectance's for each visually identified land cover class) on Landsat images with Google Earth Engine (Gorelick et al., 2017). We selected the image with the lowest cloud cover and the closest to each article's year of publication to reduce any temporal bias due to possible land-use change (see Table B.1). We classified images into three land covers: i) natural and semi-natural habitat, ii) agricultural land, and iii) other (Table B.2). The supervised classification allowed us to use high resolution (30 m) data rather than relying on existing global coarser-resolution layers (e.g., Copernicus dataset at 100 m resolution). This is important as many natural/semi-natural features in agricultural landscapes are found in small patches and agricultural fields are < 5 ha in many parts of the world (Samberg et al., 2016). We assessed the accuracy of our land cover classification by collecting samples of reflectance of ≈ 30 random points in each study area from available global or regional land cover databases (see Table B.3).

The obtained land cover data for each study area was analysed in R-4.0.0 (R Core Team, 2013) to estimate: (i) percentage of natural and semi-natural habitats (i.e., composition); (ii) land cover Shannon's diversity index (i.e., heterogeneity); and (iii) Euclidean distance from the centroid of the agricultural plot to the nearest natural or semi-natural habitat (i.e., configuration) within the 1 km radius (see description in Table B.4).

2.5. Effect size calculation

We used the weighted log-response ratio (LRR) as the effect size (Hedges et al., 1999) metric to compare the impact of diversified and simplified farming systems on non-domesticated biodiversity abundance and richness. We calculated the LRR and its variance (LRR_{VAR}) using the means, SD and number of samples provided by the included comparisons. The LRR is not biased by the number of samples when $\sqrt{\frac{n}{SD}} \times \mu$ of the interventions and comparators is ≥ 3 (Hedges et al., 1999), which was the case for 76% of comparisons in our dataset.

A mean $LRR > 0$ indicates biodiversity was positively impacted by diversified farming systems, a mean $LRR < 0$ indicates biodiversity was negatively impacted by diversified farming, and a mean $LRR = 0$ indicates no difference between diversified and simplified systems.

2.6. Meta-analysis

Most of the included articles provided multiple effect sizes corresponding to multiple: (i) time-points (i.e., articles reporting biodiversity outcomes for the same farming system in different crop stages or survey days); (ii) agricultural treatments (i.e., articles reporting biodiversity outcomes for multiple comparator-intervention systems); and (iii) outcomes (i.e., articles presenting biodiversity outcomes for multiple study sites, taxonomic or functional groups). Effect sizes from the same article can be statistically dependent, violating the assumptions of meta-regression analyses (Cheung, 2014).

The three-level meta-analysis has been proposed as a suitable approach for dealing with effect size dependency (Cheung, 2014; López-López et al., 2018; Nakagawa and Santos, 2012). In contrast to the "selecting a single effect size per article" approach, it permits the inclusion of all available information in the analysis, thus increasing statistical precision (Hedges and Pigott, 2001). The method accounts for non-independence of effect sizes within articles without requiring information on the degree of correlation between these effect sizes, information which is rarely provided at source (Van den Noortgate et al., 2013).

We conducted the meta-analyses in R-4.0.0 using the package *metafor* (v.2.4, Viechtbauer, 2010), following the approach described in Assink and Wibbelink (2016). First, we estimated the overall mean effect of diversified and simplified farming systems on species abundance and richness independently by applying an intercept only model. We followed the standard approach to applying a three-level model, accounting for random variation of the effect sizes due to sampling variance (first level); within-article variance (second level, Effect size ID); and between-article variance (third level, Article ID) (Van den Noortgate et al., 2013).

Researchers do not report species' botanical names consistently and completely. Some studies only provided general groups, such as insects or Arthropods, limiting the possibility to account for the phylogenetic relatedness of the effect sizes (e.g., Benítez-López et al., 2017). We used the t-distribution with to calculate the 95% CI for the overall mean effects (Knapp and Hartung, 2003). We used the Restricted Maximum-Likelihood method for estimating all model parameters because it is more efficient and presents less bias when calculating heterogeneity (Viechtbauer, 2005).

For abundance and species richness data, we examined the suitability of conducting the three-level meta-analysis compared to the two-level model using one-sided log-likelihood-ratio tests for the variance at the second and third levels separately (Assink and Wibbelink, 2016). The three-level model is justified when variances at the first second and third level are significantly greater than zero (Cheung, 2014). In addition to the heterogeneity parameters tested by the model, we quantified the proportion of the observed variance (I^2) for each of the three levels ($I^2_{(1)}$, $I^2_{(2)}$, $I^2_{(3)}$) using the formulas provided in Cheung (2014). The application of moderator analyses is justified when the $I^2_{(1)}$ is low (near

zero) compared to the total variance (Borenstein et al., 2009).

We applied univariate meta-regression procedures to examine whether the functional groups (categorical variable with 6 classes), and landscape composition (continuous variables), heterogeneity (continuous variables), and configuration (categorical variables with 6 classes) metrics moderated the effect of farming systems on biodiversity. Landscape configuration (i.e., Euclidean distance from the centroid of the agricultural plot to the nearest natural or seminatural habitat), was categorised for capturing differences at smaller and larger ranges beyond the 1 km threshold. An omnibus test based on the F-distribution was used to evaluate whether there were significant differences in effect sizes across the values of each moderator (Viechtbauer, 2010). We also examined if the influence of each landscape metric on the estimated effect size were the same across the six functional groups, by including the interaction between functional groups and each landscape metric as a moderator in the meta-regression models.

To facilitate the interpretation of the results, we reported the mean effect sizes and their CI as percentages changes by back-transforming the log response ratios and converting them to percentages ($100 \times (\exp(LRR) - 1)$) (Pustejovsky, 2018). A mean effect size > 0 indicates biodiversity was positively impacted by diversified farming, whereas a mean effect size < 0 indicates biodiversity was negatively impacted by diversified farming. A mean effect size equal to zero indicates little or no difference between diversified and simplified systems.

We applied multivariate models (i.e., the inclusion of multiple moderators in the same model) to check which variables continued significantly moderating the overall effect when controlling for other variables (Ho, 2006). The inclusion of highly collinear moderators (i.e., predictors sharing considerable amounts of information) in the same model, however, may lead to problems of overfitting or spurious results (Dormann et al., 2013; Ho, 2006). We tested for multicollinearity between moderators by computing the Cramer's V correlation between categorical variables, Spearman correlation between continuous variables, and Kruskal-Wallis H test between categorical and continuous variables. We considered moderators as presenting severe collinearity between each other when Cramer's V or Spearman correlation were ≥ 0.7 or Kruskal-Wallis H tests were significant (Akoglu, 2018; Dormann et al., 2013) (see Table A.4). We constructed separated multivariate models for abundance and species richness by adding all non-collinear moderators found to be significant predictors during the univariate models. We ignored collinearity between functional groups and the landscape metrics, since they are conceptually very different (Dormann et al., 2013). We used omnibus tests to compare the significance of each moderator that had been found to be significant in the univariate models.

Finally, we used multi-model inference to check the model fit statistics of each applied model based on the corrected Akaike's Information Criterion (AICc) (see Table A.5). We used the AICc as our selection method because it is considered to be less biased and selects models that are much closer to the truth (Burnham and Anderson, 2002).

2.7. Publication bias analysis

Publication bias may occur due to the inability to identify all articles that addressed the research question (Borenstein et al., 2009). Following the statistical publication bias analysis for databases with non-independent effect sizes (Kunc and Schmidt, 2019; Tamburini et al., 2020), we conducted the Egger's regression test including the inverse of the standard error of the effect sizes as a moderator. If the intercept obtained from the regression test is significantly different from zero at a 95% CI, the database presents evidence of publication bias (Egger et al., 1998). Furthermore, we used funnel plots as a visual approach to detect publication bias in the abundance and richness databases (Borenstein et al., 2009).

Some meta-analyses have also detected evidence of bias caused by temporal and geographical differences across effect sizes (Shackelford

et al., 2013; Tuck et al., 2014). We examined whether effect sizes vary across time and location by including publication year and latitude as moderators in separate meta-regression models for species abundance and richness (Nakagawa and Santos, 2012). If the results from the omnibus tests are significant, it can be assumed that the effect sizes differ across year and location.

2.8. Sensitivity analysis

Results from meta-analysis can be influenced by outliers (i.e., highly influential effect sizes) in the data, compromising the robustness of the conclusions. Following Viechtbauer and Cheung (2010), we defined potential outliers as effect sizes with Cook's distance values (calculated using the *cook.distance* function, Viechtbauer, 2010) greater than the chi-squared distribution with $df = (k + 1)$ and $\alpha = 0.5$, or comparatively larger when compared to the other effect sizes.

We explore the influence of the effect sizes outliers and effect sizes with high and unknown risk of bias, by running the intercept and meta-regression models without these effect sizes for the species abundance and richness analyses (Aguinis et al., 2013). If the exclusion of potentially biased or outlier effect sizes did not change the directionality of the mean values and the statistical significance of the meta-analysis results, then the results were considered robust.

3. Results

3.1. Meta-data description

The majority of data in our dataset represented Arthropods (1585 (84%) effect sizes for abundance and 192 (51%) effect sizes for species richness) and the majority of these were classified as crop pests and pollinators for species abundance and richness effect sizes, respectively (Table A.6). Most of the abundance data were collected from associated plant species (665 effect sizes), and intercropping farming systems (639 effect sizes), while most of the species richness data came from agroforestry (150 effect sizes), embedded natural (64 effect sizes), and crop rotation (45 effect sizes) farming plots (see Table A.7).

Although our dataset is extensive, effect sizes were unevenly distributed across space, with North America providing 40% of the abundance data and Africa providing no data on species richness (Fig. A3). Our meta-data included articles that were published between 1986 and 2021, but 89% and 98% of the articles for abundance and species richness, respectively, were published after 2000 (Fig. A.4). In terms of quality, over half of the effect sizes for species abundance (59%) and richness (53%), presented low risk of bias (see Fig. A.5).

3.2. Effects of farming systems on abundance

Diversified farming systems had a positive but non-significant impact on abundance, increasing the (back-transformed) overall mean effect size by 7% [95% CI = $-8, 25$] compared to simplified systems (Fig. 1). The heterogeneity analyses revealed that 99.9% of the total variance corresponds to effect sizes differences within and between articles (Table A.8), justifying the application of meta-regression analyses.

Univariate meta-regression showed that the impact of farming systems on abundance varies significantly among functional groups ($F_{(6, 1880)} = 14.568, p < 0.001$). Pollinators and decomposers responded positively to diversified farming, while autotrophs and natural enemies showed a positive but not significant response (Fig. 1). Conversely, the abundance of pests was significantly lower (-21% [95% CI = $-32, -8$]) in diversified farming systems than in simplified systems.

The impact of diversified farming systems on abundance was not moderated by the percentage of natural and semi-natural habitats ($F_{(1, 1884)} = 0.089, p = 0.765$), land cover Shannon's diversity index ($F_{(1, 1884)} = 0.451, p = 0.502$), or distance to the nearest natural or semi-natural habitat in 1 km radii ($F_{(6, 1880)} = 1.412, p = 0.206$) (see

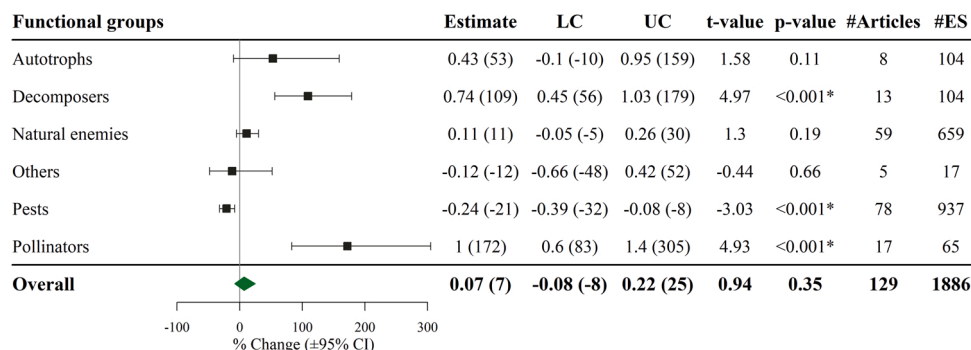


Fig. 1. The effects of farming systems (diversified vs. simplified) on organismal abundance in six functional groups and overall. Estimate = mean effect sizes in LRR (values in percentage of change); LC = lower confidence limit; and UC = upper confidence limit at 95% in LRR (values in percentage); #Articles = number of included articles; #Effect sizes = number of included effect sizes; *Mean effect size significantly different from zero $\alpha = 0.05$.

Table A.9; Fig. A.6).

We found significant interaction effects between functional groups abundance and the landscape metrics (Table A.10). The positive impact of diversified farming systems on decomposers abundance increased with the increment of the percentage of natural habitats, land cover diversity, and when plots were near (50–100 m) and very far away (>1000 m) from natural habitats (Fig. 2). The impact of diversified farming systems on natural enemies' abundance were significantly positive in farming plots located near (50–100 m) and moderately near (100–250 m) to natural habitats. Contrastingly, the abundance of pests was significantly reduced in diversified farming systems located very

close (0–50 m) or moderately near (100–250 m) to natural habitats, and in landscape with high land cover diversity. The abundance of pollinators was significantly higher in diversified farming systems located moderately far (250–500 m) and very far away (>1000 m) from natural habitats, and in agricultural landscape with high land cover diversity.

The omnibus tests from the multivariate models applied (i.e., separate models adding the non-collinear moderators that were significant during the univariate analyses) for abundance showed that functional groups and the interaction between Functional groups and Euclidean min-distance to natural or semi-natural habitats remained significant

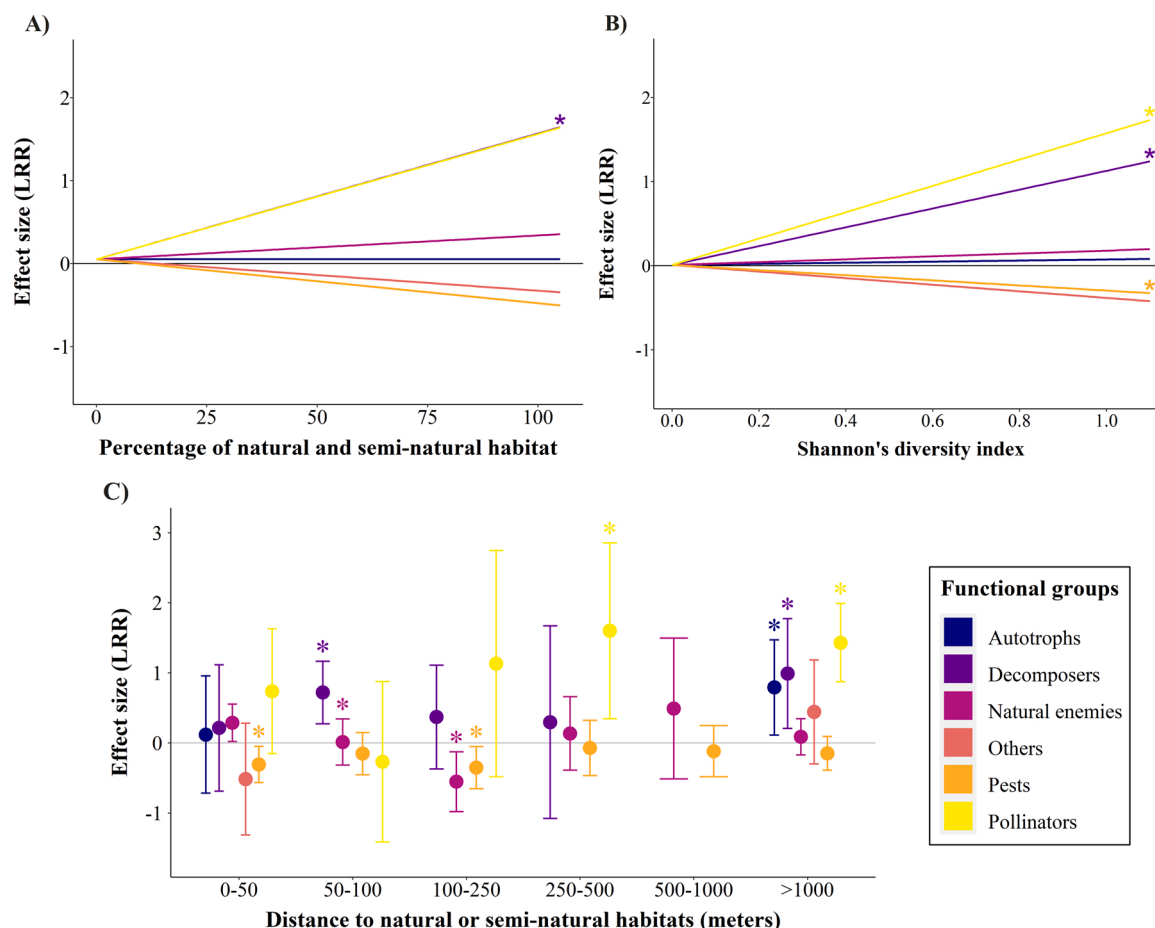


Fig. 2. Mean effect size of farming systems (diversified vs. simplified) on abundance from the interaction between functional groups and: A) the percentage of landscape covered by natural and semi-natural habitats; B) land cover Shannon's diversity index; C) categorical variables for the nearest Euclidean distance to natural or semi-natural habitat. Lines and dots = mean effect sizes in LRR. Error bars = $\pm 95\%$ CI. *Mean effect size significantly different from zero $\alpha = 0.05$.

moderators (Table A.11). However, the interaction effects between Functional groups and Percentage of natural habitat, and Functional groups and Land cover Shannon's diversity index were no longer significant when including Functional groups in the models.

3.3. Effects of farming systems on species richness

Diversified farming systems had, on average, 26% [95% CI = 13, 40] higher species richness than in simplified systems (Fig. 3). Heterogeneity analyses supported the application of three-level meta-analysis and meta-regression models (Table A.12).

As with abundance, the impact of farming systems on species richness varied significantly among functional groups ($F_{(6, 366)} = 5.156$, $p < 0.001$). However, the effect tended to be positive for both beneficial and non-beneficial groups (Fig. 3). The richness of autotrophs, natural enemies, and pollinators were significantly higher in diversified farming systems, and that of decomposers and pests non-significantly higher. Species richness for the 'others' functional group was non-significantly lower.

For landscape composition, the mean effect of diversified farming systems on species richness significantly decreased by $\approx 0.4\%$ [95% CI = $-0.9, -0.017$] with every 1% increase in the percentage of natural and semi-natural habitats in a 1 km radius. Moreover, the overall mean effect of diversified farming on species richness significantly increased by 33% [95% CI = 6, 68] in agricultural plots located moderately near (100–250 m) and 41% [95% CI = 18, 68] in plots located very far away (> 1000 m) from natural or semi-natural habitats. Landscape heterogeneity did not moderate the impact of diversified farming systems on species richness (see Fig. 4, Table A.13).

The positive effect of diversified farming systems on autotrophs and pollinator richness increased with increasing land cover diversity. The positive effect on the richness of both pollinators and natural enemies was highest when the agricultural plot was very far (> 1000 m) from natural and semi-natural habitats. Farming diversification had a significant positive effect on autotrophs richness when increasing the percentage of natural and semi-natural habitats around and in plots located near (50–100 m) and moderately near (250–500 m) (see Fig. 5, Table A.14).

The test of moderators showed that none of the moderators found to be significant predictors in the univariate models remained significant when controlling for effects of the other included variables (Table A.15).

3.4. Publication bias analysis

For the abundance database, the adapted Egger's regression test ($t_{1884} = 1.191$, $p \sim 0.234$) (Table A.16a) and the funnel plot (Fig. A.7a) revealed no evidence of publication bias. The meta-regression models revealed that the mean effect size did not vary across the year of

publication of articles and the geographic location of the studies (Table A.17a).

For the species-richness database, the intercept from the Egger's regression test was significant ($t_{370} = 3.46$, $p = 0.0006$) (Table A.16b), and the visual inspection showed clear asymmetry (Fig. A.7b), suggesting evidence of publication bias. The meta-regression models revealed that the mean effect size significantly varied across the articles' publication year with an annual decrease of 2.2% (Table A.17b). Whereas, the geographic location of the studies did not moderate diversified farming systems effect on species richness.

3.5. Sensitivity analysis

The identification of effect sizes outliers for the abundance and species richness models, revealed that some effect sizes presented Cook's distances values comparatively larger than others (Fig. A.8).

The sensitivity analyses exposed that the directionality and significance of the overall mean effect of farming systems on abundance were unaffected by the exclusion of effect sizes outliers (9% [95% CI = $-6, 27$]; $p = 0.24$) and effect sizes with high and unknown risk of bias (16% [95% CI = $-6, 27$]; $p = 0.16$). Diversified farming systems had a significant positive impact on autotrophs after removing the effect sizes outliers (95% [95% CI = 16, 227]; $p = 0.01$) and with high and unknown risk of bias (184% [95% CI = 33, 506]; $p = 0.01$) from the functional group meta-regression models, while the mean direction and significance of the other groups were unaffected (Fig. A.9). The significance of the moderating effects of the three landscape metrics did not vary with the exclusion of effect size outliers or effect sizes with high or unknown risk of bias (Table A.18).

Diversified farming systems had a weak positive effect on the overall mean species richness, after excluding effect size outliers (32% [95% CI = $-2, 78$]; $p = 0.07$). However, excluding the effect sizes with high and unknown risk of bias did not change the direction or significance of the overall mean effect for species richness (27% [95% CI = 12, 43]; $p < 0.001$) (Fig. A.10). The effect size direction and significance of species richness across functional groups did not vary after excluding outliers from the model. However, after removing the biased effect sizes, diversified farming systems had a significant positive impact on decomposers richness (31% [95% CI = 2, 70]; $p = 0.04$), while the positive effect on pollinators was no longer significant (24% [95% CI = $-4, 60$], $p = 0.09$). The results from the univariate meta-regressions with landscape composition, heterogeneity, and configuration metrics as moderators did not change in direction or significance after excluding the outliers and potentially biased effect sizes (Table A.19).

4. Discussion

This meta-analysis showed that, on average, diversified farming

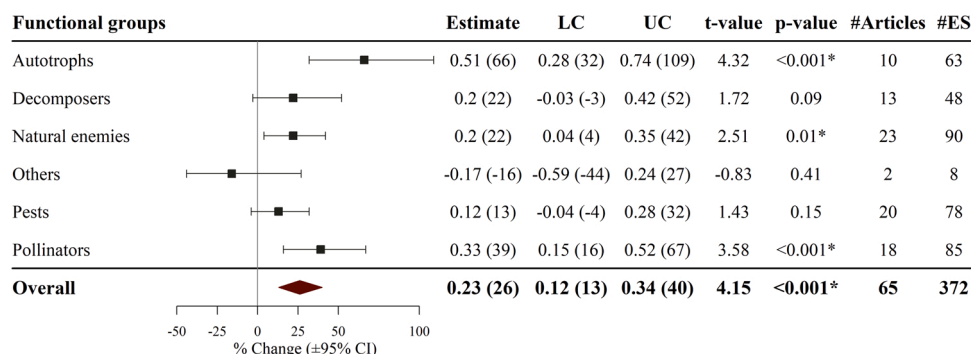


Fig. 3. The effects of farming systems (diversified vs. simplified) on species richness in six functional groups and overall. Estimate = mean effect sizes in LRR (values in percentage of change); LC = lower confidence limit and UC = upper confidence limit at 95% in LRR (values in percentage); #Articles = number of included articles; #Effect sizes = number of included effect sizes; *Mean effect size significantly different from zero $\alpha = 0.05$.

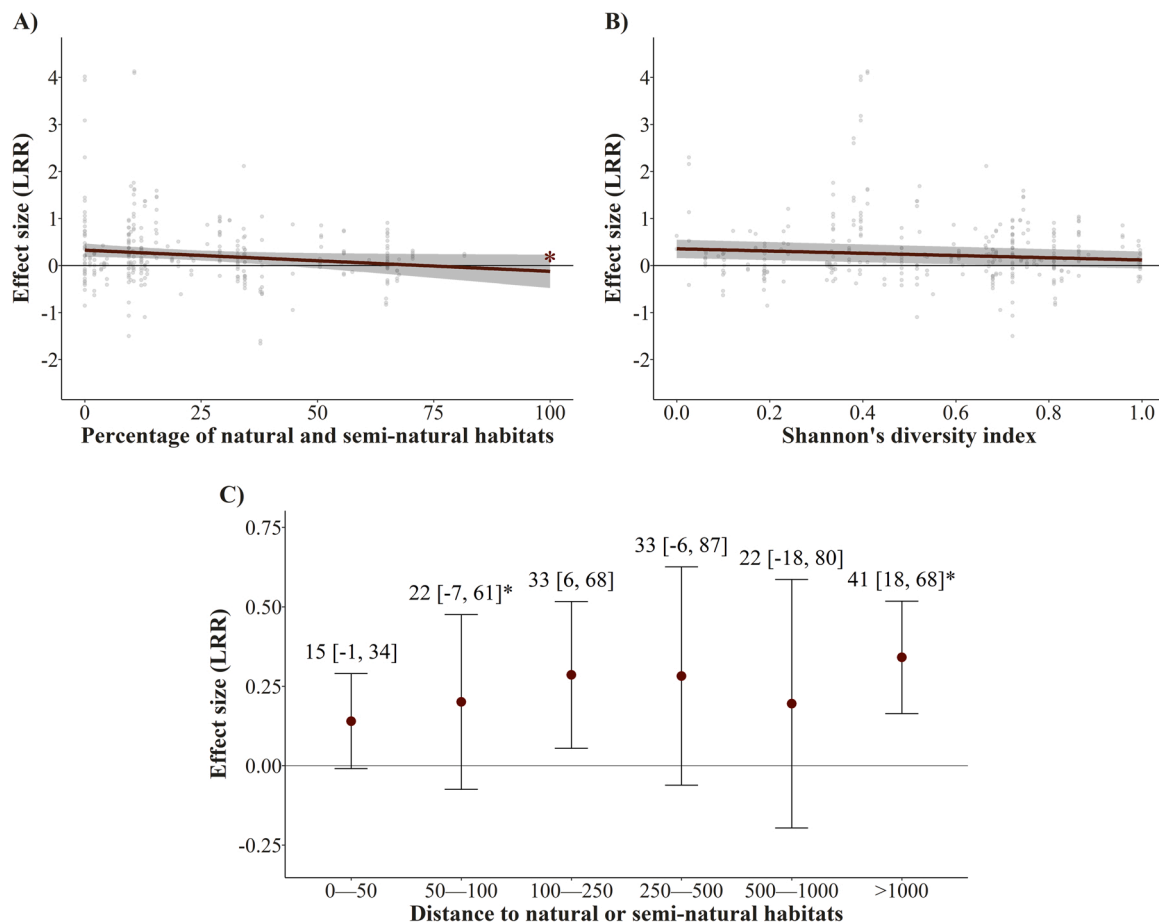


Fig. 4. Mean effect size of farming systems (diversified vs. simplified) on species richness in relation to: A) the percentage of landscape covered by natural and semi-natural habitats B) land cover Shannon's diversity index; C) categorical variables for the nearest Euclidean distance to natural or semi-natural habitat (labels = the mean effect size in percentage of change and the $\pm$ 95% CI). The red line and red dots = the mean effect sizes in LRR. Grey shadow and error bars = $\pm$ 95% CI. Grey points = the included effect sizes. *Mean effect size significantly different from zero $\alpha = 0.05$.

systems improves the richness and abundance of non-domesticated taxa with potential benefits for local and global biodiversity and food production goals. The magnitude of the overall effect varies with functional group and landscape complexity, heterogeneity and composition.

Similar to [Lichtenberg et al. \(2017\)](#), our study showed that agricultural diversification substantially increased species richness, and had a non-significant positive effect on abundance. This may reflect that farming diversification helps support a wider range of species, probably as more diverse habitat and resources open up ([Kremen, 2020](#); [Kremen and Merenlender, 2018](#)). However, diversification may limit simultaneously individual species dominance (i.e., the widespread of individual dominant species) reducing the population numbers for some species (mainly pests) ("enemies' hypothesis" - [Root, 1973](#)). For example, our results show that while the abundance of beneficial functional groups tended to have positive effect sizes, pest populations significantly decreased in diversified systems and therefore reduced the overall mean effect size for abundance.

The variable effects of diversified farming systems across functional groups supports the notion that different species respond to biotic interactions and abiotic conditions according to their functional traits ([Suding et al., 2008](#)). The negative effect of diversified farming systems on pest abundance may be related to higher predation and parasitism rates in diversified farming systems (enemies' hypothesis; [Letourneau et al., 2011](#); [Root, 1973](#)), due to synergistic effects between species (e.g., complementarity and facilitation in the partitioning of habitat and resources; and reduction of the competitive exclusion among species) ([Hooper et al., 2005](#)). Indeed we found a higher richness of natural

enemies in diversified systems, consistent with several previous meta-analyses ([Greenop et al., 2018](#); [Dainese et al., 2019](#)). Our study extends on previous work by including pest plants (weeds) and our findings suggest the pest control benefits of diversified farming systems extend to weed suppression. Crop and farm diversification may be an effective weed management strategy helping to reduce the need for herbicide inputs ([Petit et al., 2020](#)).

Our synthesis demonstrated pollinator richness and abundance benefited from diversified farming practices, consistent with previous quantitative syntheses ([Kennedy et al., 2013](#); [Lichtenberg et al., 2017](#)). The positive effect of diversified farming systems on the concentration of pollinator species might lead to a greater provision of pollination services with benefits to crop yields ([Dainese et al., 2019](#); [Suding et al., 2008](#)). Ensuring pollinator abundance may be more important than richness for provision of pollination services since pollination in agroecosystems might be mostly delivered by a few abundant and widespread species of insects rather than rare pollinator species ([Kleijn et al., 2015](#)). Nevertheless, a high redundancy of pollinators can help ensure community stability and function provision against unexpected changes ([Schleuning et al., 2015](#)).

Agroecosystem resilience is associated with high diversity of organisms responsible for maintaining soil structure and nutrients cycling ([Swift et al., 2004](#)). Conversely to previous studies that indicated local management had little effect on decomposers (mainly Arthropods) ([Lichtenberg et al., 2017](#); [Tuck et al., 2014](#)), our results – including a wider range of taxa (i.e., Annelids, Arthropoda Eukaryote, and Fungi) – demonstrated that diversified farming systems substantially increased

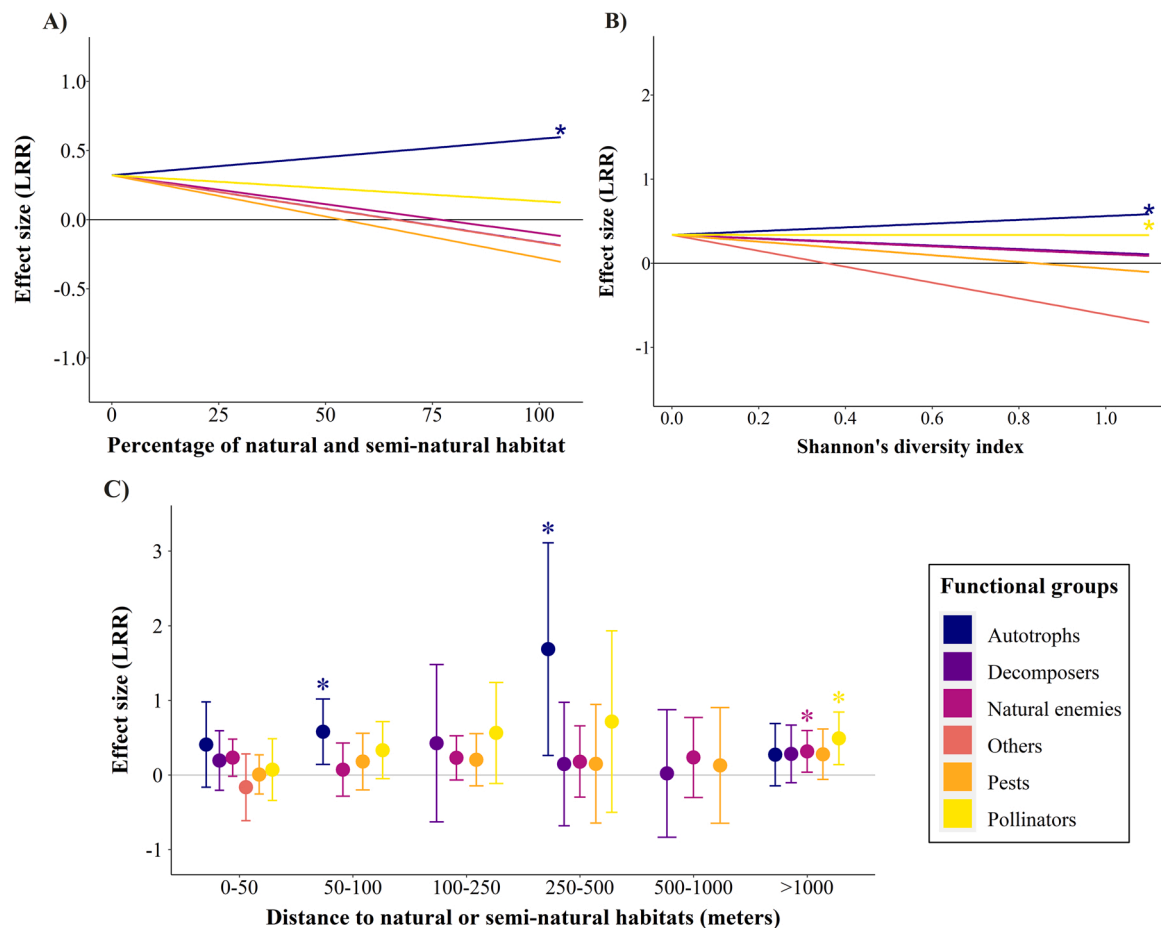


Fig. 5. Mean effect size of farming systems (diversified vs. simplified) on species richness from the interaction between functional groups and: A) the percentage of landscape covered by natural and semi-natural habitats; B) land cover Shannon's diversity index; C) categorical variables for the nearest Euclidean distance to natural or semi-natural habitat. The lines and dots = mean effect sizes in LRR. Error bars = $[\pm 95\% \text{ CI}]$. *Mean effect size significantly different from zero $\alpha = 0.05$.

decomposer abundance. This may be because diversified farming systems have a higher concentration of soil organic matter, benefiting soil biota (Bender et al., 2016). Effects on decomposers may also be associated with changes in autotrophs (non-pest plants) identified in this study, since a higher richness of autotrophs can supply more diverse resources (e.g., plant litter, root configuration) which may enhance the decomposer community (Beare et al., 1997).

Simpler landscape configurations enhanced the positive effect of diversified farming systems on the overall biodiversity richness in our study, specifically when natural habitats were > 1000 m away or constituted $\approx < 60\%$ of the surrounding 1 km radii area. Our results agree with the "intermediate landscape complexity hypothesis", which establish the positive effect of local-conservation practices might be higher in agricultural plots located in simple landscapes than in complex ones (Tschamtko et al., 2012). This response may be because in landscapes with a lower proportion of natural habitats, diversified farming systems (e.g., presence of hedgerows, flower vegetation strips) may offer more varied habitats (e.g., for nesting, and refuge) and resources (e.g., food, pollen) attracting a higher variety of organisms than monocultures (Kremen and Merenlender, 2018). Hence, diversification strategies at the farm level should complement restoration approaches at the landscape scale (e.g., increase the proportion of natural habitats) (Garibaldi et al., 2020) for promoting biodiversity, and softening productivity-ecosystem services trade-offs (Tamburini et al., 2020).

Similar to prior observations (Karp et al., 2018; Kennedy et al., 2013), we found some interaction effects between functional groups diversity and the surrounding landscape characteristics. Our synthesis showed the importance of farming diversification at increasing

pollinators abundance and richness in highly simplified and heterogeneous landscapes. Farm-scale diversification strategies might counter the direct negative impact of the absence of surrounding natural or semi-natural habitats on pollinators number and diversity, and therefore, enhance pollination service provision (Dainese et al., 2019). Natural and semi-natural habitats close to farming plots might be important to sustain populations of predators which benefit from the resources offered by natural habitats (e.g., shelter, nest sites) as well as by adjacent crops (e.g., pollen, prey) (Martin et al., 2019). This may also explain the reduction in pest abundance in diversified farms close to natural and semi-natural habitats, where predators were more abundant helping limit infestations (Gontijo, 2019).

Moreover, our results showed the positive effects of diversified farming systems on decomposer abundance tend to increase with landscape complexity, primarily the proportion of surrounding natural habitats and landscape heterogeneity, but also with the distance from natural and semi-natural habitats. These results suggest the colonisation of soil biota may depend on the surrounding landscape and organisms' dispersal abilities (Auclerc et al., 2009). However, decomposer richness seems to be independent of landscape patterns (Schneider et al., 2007), and more influenced by local characteristics such as soil type, climate, plant diversity, and temperature (Tschamtko et al., 2012). On the other hand, autotrophs (plants) responded positively to farming system diversification as landscape complexity increased, which may reflect that higher landscape complexity facilitates seed pool dispersal and establishment (Kirmer et al., 2008). Besides, most of the autotrophs' data came from the comparison effect of simplified and agroforestry systems, suggesting the establishment and development of non-crop

plants also could be promoted by the lower herbicide input or soil disturbance practices that characterised these diversified systems (Boinet et al., 2019).

Results from the multivariate models revealed the robustness of the unique moderating effect of functional groups, and the interaction between landscape configuration and functional groups, on abundance. However, the interaction between landscape composition and heterogeneity with functional groups were no longer significant when accounting for other variables in the same model. These results may be because landscape composition and heterogeneity interact with functional groups in determining effect sizes, but functional group has a stronger effect on its own. For species richness, none of the non-collinear predictors with significant moderating effects within the univariate models remained significant when included together in the same model. The reduction in the significance of these variables may be related to the lower number of effect sizes in the species richness database ($n = 372$), which leads to small class sizes when multiple variables are combined in a single model. When the sample size is small, multivariate models may be statistically less powerful than univariate models (Maxwell et al., 2017). While our results show that functional group, and distance to and proportion of natural and semi-natural habitat, moderate the effect of diversified farming on species richness, future research to expand the number of effect sizes in the richness database as new studies emerge would be valuable to confirm which moderators have the strongest influence.

The sensitivity and publication bias analysis, in addition to the pre-publication of the review protocol (Sánchez et al., 2021) and the meta-data (Jones et al., 2021), guarantee the transparency and reproducibility of our meta-analysis (Beillouin et al., 2019). While sensitivity analyses confirmed the robustness of our overall findings for abundance, we identified some changes in the significance of the species richness results when excluding the effect size outliers and with a high risk of bias from the analyses. The exclusion of one highly influential effect size reduced the significance of the overall farming systems impact on species richness by widening its confidence interval. However, the significance of the overall species richness results was not affected by the exclusion of 175 effect sizes with a high risk of bias. Hence, this last finding may ensure the robustness of our overall results for species richness. On the other hand, the reduction in significance of positive effect of diversified systems on pollinators species richness may be related to the exclusion of almost 40% of the effect sizes. Moreover, we identified possible sources of publication bias that may reduce the generalisation of our conclusions. However, the decrease in overall effect size on biodiversity with increasing publication year may not be a reflection of research bias, and rather a reflection of real changes in functional diversity of non-domesticated taxa observed on the ground. On the one hand, the reduction in the positive effect of diversified farming systems on species richness across time may be directly related to the unprecedented biodiversity loss terrestrial systems experienced in recent decades (WWF, 2020). On the other hand, over the years, diversified farming systems may provide shelter and the necessary resources to support more abundance, especially given the increasing reduction and degradation of natural habitats in agricultural landscapes (Laurance et al., 2014).

4.1. Limitations and future work

Compared to previous syntheses, our meta-analysis combined a larger number of articles and effect sizes, being also the first to include other taxa in addition to arthropods. However, there were some geographic and taxonomic gaps in our database (e.g., species-richness data in Africa and abundance data for amphibians and reptiles) that should be considered when generalising results. The next step for meta-analyses in this topic, could be to include primary articles in languages other than English and/or unpublished/empirical data to cover poorly represented geographic areas and taxonomic groups. Additionally,

further research measuring and quantifying wild biodiversity across multiple production systems with different diversification strategies and landscapes with varying landscape complexity is highly needed.

Our landscape analysis used an innovative and replicable methodology to classify land-cover, while limiting the potentially confounding effect of land-use change over time, and using high-resolution data that is particularly appropriate where smallholdings are widespread (e.g., in Africa, Asia, Central and South America). However, it was difficult and time-consuming to find consistent training data across countries.

Although our results demonstrate the ecological and agronomical benefits diversified farming systems could have, the decision of farmers to adopt these practices is limited by some uncertainties about how to apply them (Duru et al., 2015). Similarly, the potentially associated socio-economic challenges (e.g., higher investments or labour costs, market or policy conditions that favour simplify agricultural systems) may limit farmers' adoption of diversified farming systems (Kremen and Merenlender, 2018; Rosa-Schleich et al., 2019). Hence, more studies that connect scientific and empirical knowledge of ecological and agronomic processes in diversified farming are needed to promote the effective and viable implementation of these systems by farmers across regions. Future meta-analyses should, therefore, cover the agronomic cost-benefits of the application of diversified and simplified farming systems.

5. Conclusion

Our meta-analytical method confirms farm diversification contribute directly or indirectly to production objectives by increasing the abundance or richness of beneficial organisms, such as pollinators, natural enemies, and decomposers, and reducing the abundance of pest species. However, ensuring also the greatest benefit to biodiversity requires considering the landscape context. The multiple benefits of farm diversification such as improving small farmers' income, ensuring food and nutrition security year-round, as well as promoting the conservation of natural resources and biodiversity are of particular importance in developing countries where farm simplification and expansion is taking place at a rapid pace (Altieri et al., 2012). Finally, our results contribute clear evidence of agricultural land's role to halt biodiversity loss, if properly managed. This information is very timely, to inform current global agendas such as the Convention on Biological Diversity COP15 where policies, regulations, and international agreements can be established and repurposed to incentivise the adoption and spread of farm and landscape diversification.

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Authors contribution

All authors read and approved the final version of the manuscript. A.C.S., S.K.J. and N.E.C. conceived of the study. A.C.S. designed the methodology, conducted the analysis, and wrote the first draft of the manuscript. A.D.P., A.P., S.K.J., N.E.C. reviewed and edited the manuscript. A.D.P. supervised the correct implementation of the methodology and interpretation of the results.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data linking: <https://doi.org/10.7910/DVN/KWGD8S>.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.agee.2022.107933](https://doi.org/10.1016/j.agee.2022.107933).

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